Introduction to NumPy arrays

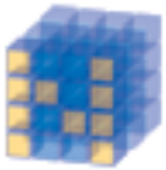
https://github.com/gertingold/euroscipy-numpy-tutorial.git

Python comes with batteries included

➜ extensive Python standard library

What about batteries for scientists (and others as well)?

➜ scientific Python ecosystem



NumPy

Base N-dimensional array package

SciPy library

Fundamental library for scientific computing

Matplotlib

Comprehensive 2D Plotting

IP[y]: IPython

IPython

Enhanced Interactive Console

Sympy

Symbolic mathematics

pandas

Data structures & analysis

from: www.scipy.org

+ SciKits and many other packages

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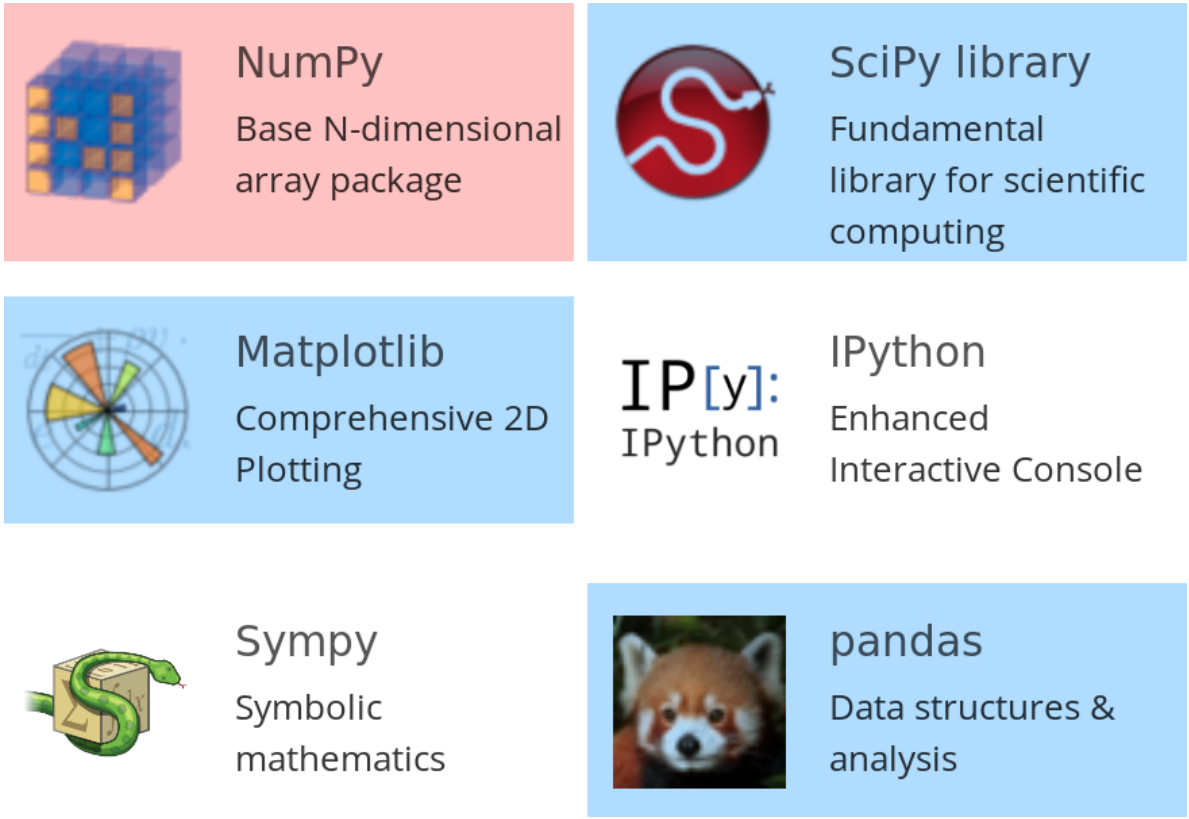

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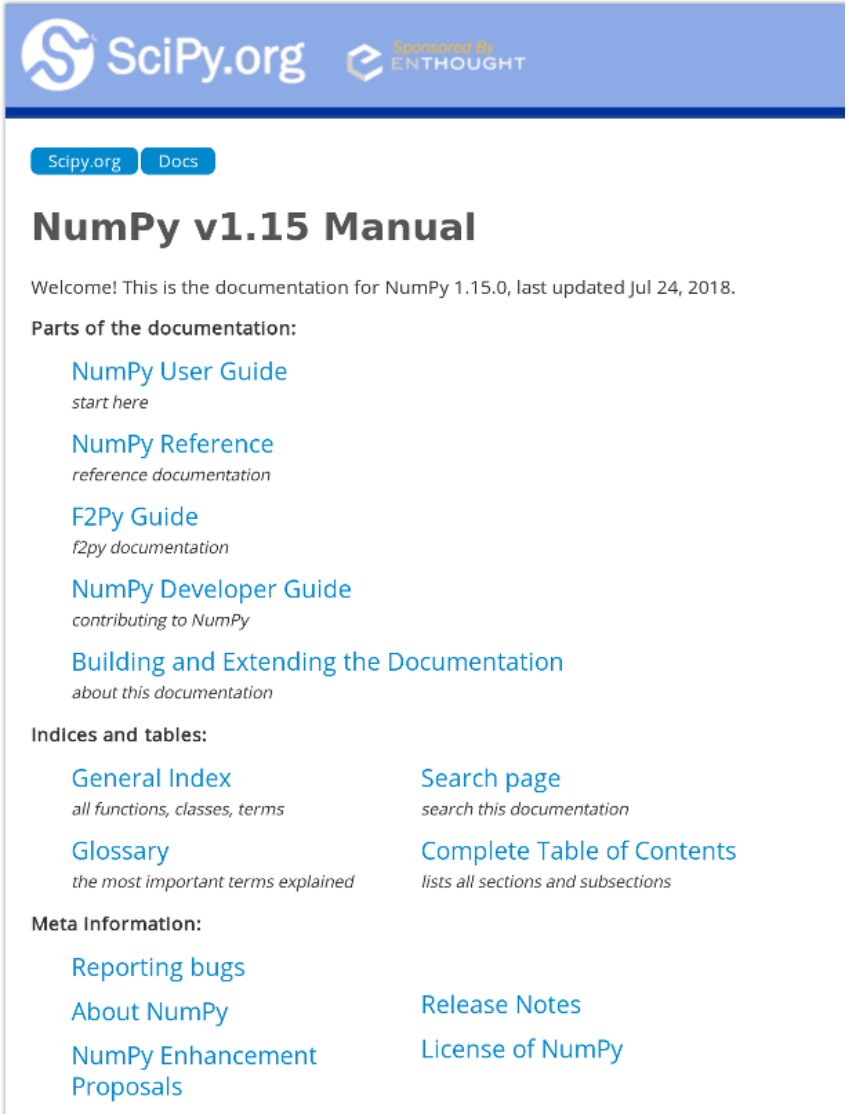
www.scipy-lectures.org

docs.scipy.org/doc/numpy/



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Scipy.org Docs

NumPy v1.15 Manual

Welcome! This is the documentation for NumPy 1.15.0, last updated Jul 24, 2018.

Parts of the documentation:

NumPy User Guide
start here

NumPy Reference
reference documentation

F2Py Guide
f2py documentation

NumPy Developer Guide
contributing to NumPy

Building and Extending the Documentation
about this documentation

Indices and tables:

General Index
all functions, classes, terms

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search this documentation

Glossary
the most important terms explained

Complete Table of Contents
lists all sections and subsections

Meta Information:

Reporting bugs

About NumPy

Release Notes

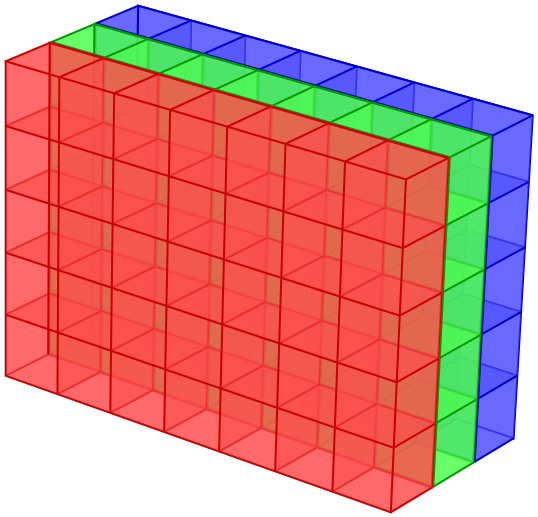
NumPy Enhancement Proposals

License of NumPy

A wish list

- we want to work with vectors and matrices

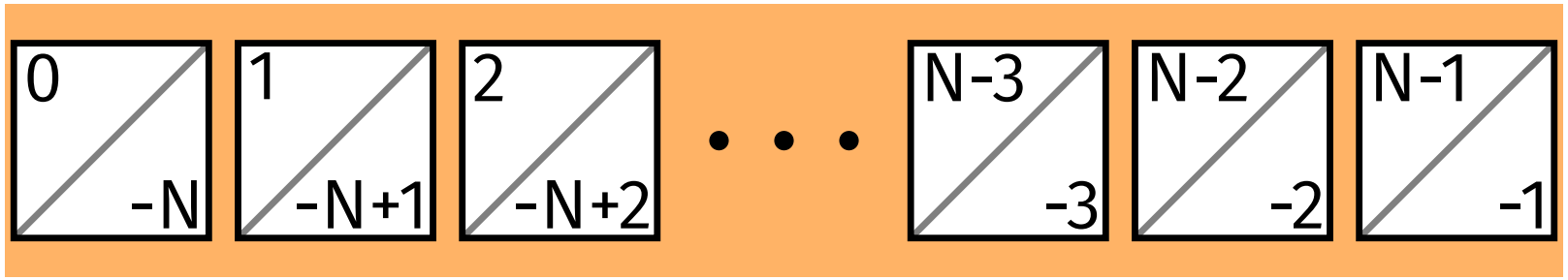
$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}$$



colour image as $N \times M \times 3$-array

- we want our code to run fast
- we want support for linear algebra
- ...

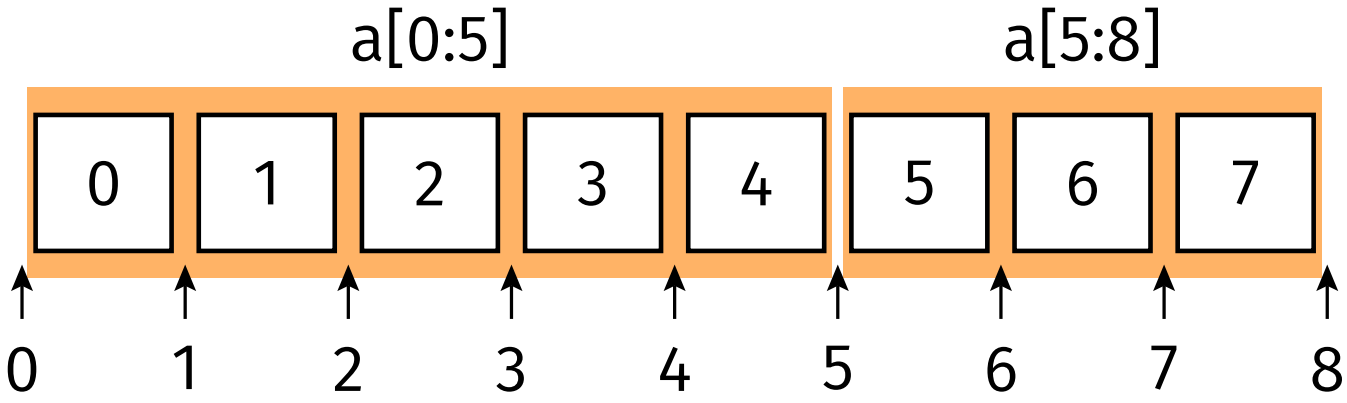
List indexing



- indexing starts at 0
- negative indices count from the end of the list to the beginning

List slicing

basic syntax: `[start:stop:step]`



- if `step=1`
  - slice contains the elements `start` to `stop-1`
  - slice contains `stop-start` elements
- `start`, `stop`, and also `step` can be negative
- default values:
  - `start` 0, i.e. starting from the first element
  - `stop` N, i.e up to and including the last element
  - `step` 1

Let’s do some slicing

Matrices and lists of lists

Can we use lists of lists to work with matrices?

$$\begin{pmatrix} 0 & 1 & 2 \\ 3 & 4 & 5 \\ 6 & 7 & 8 \end{pmatrix}$$

```
matrix = [[0, 1, 2],
          [3, 4, 5],
          [6, 7, 8]]
```

- How can we extract a row?
- How can we extract a column?

Matrices and lists of lists

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```

- How can we extract a row?
- How can we extract a column?

Let's do some experiments

Matrices and lists of lists

Can we use lists of lists to work with matrices?

$$\begin{pmatrix} 0 & 1 & 2 \\ 3 & 4 & 5 \\ 6 & 7 & 8 \end{pmatrix}$$

```
matrix = [[0, 1, 2],
          [3, 4, 5],
          [6, 7, 8]]
```

- How can we extract a row? ☺
- How can we extract a column? ☹

Lists of lists do not work like matrices

Problems with lists as matrices

- different axes are not treated on equal footing
- lists can contain arbitrary objects
  matrices have a homogeneous structure
- list elements can be scattered in memory

Applied to matrices ...

...lists are conceptually inappropriate

...lists have less performance than possible

We need a new object

ndarray

multidimensional, homogeneous array of fixed-size items

Getting started

Import the NumPy package:

```
from numpy import *
```

Getting started

Import the NumPy package:

~~`from numpy import *`~~

```
from numpy import array, sin, cos
```

Getting started

Import the NumPy package:

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import numpy
```

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~~`import numpy`~~
`import numpy as np` ←

Getting started

Import the NumPy package:

~~`from numpy import *`~~
~~`from numpy import array, sin, cos`~~
~~`import numpy`~~
`import numpy as np` ←

Check the NumPy version:

```
np.__version__
```

Data types

Some important data types:

integer	`int8`, `int16`, `int32`, `int64`, `uint8`, ...
float	`float16`, `float32`, `float64`, ...
complex	`complex64`, `complex128`, ...
boolean	`bool8`
Unicode string	

default type: `float64`

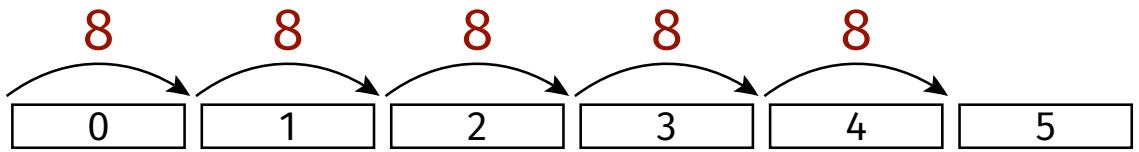
Beware of overflows!

Your turn

Strides

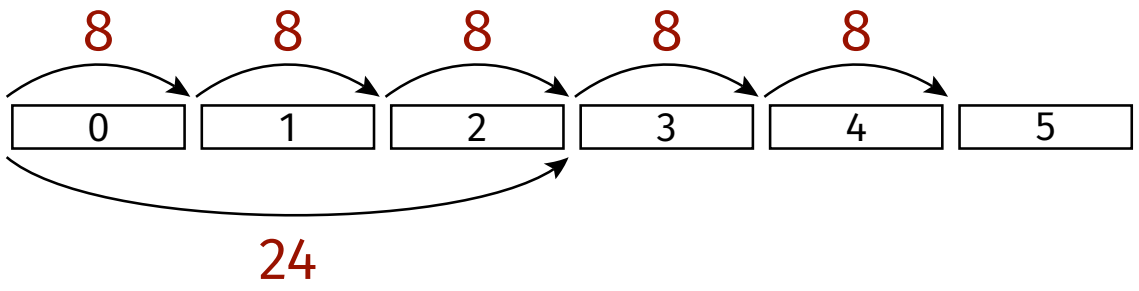
$$\begin{pmatrix} 0 & 1 & 2 & 3 & 4 & 5 \end{pmatrix}$$

(8,)



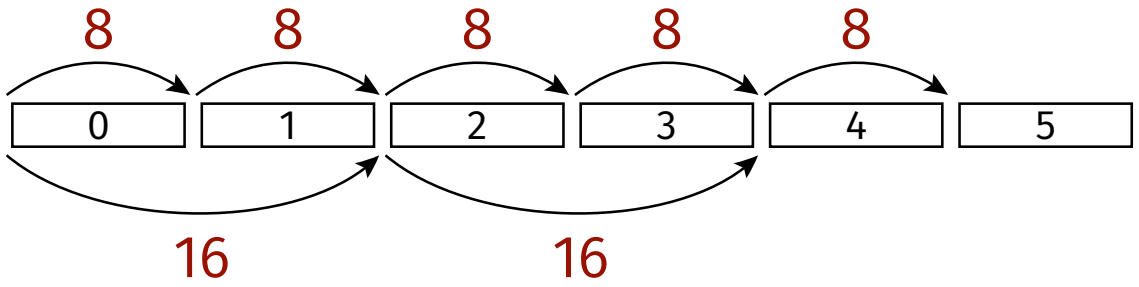
$$\begin{pmatrix} 0 & 1 & 2 \\ 3 & 4 & 5 \end{pmatrix}$$

(24, 8)



$$\begin{pmatrix} 0 & 1 \\ 2 & 3 \\ 4 & 5 \end{pmatrix}$$

(16, 8)



Views

For the sake of efficiency, NumPy uses views if possible.

- Changing one or more matrix elements will change it in all views.
- Example: transposition of a matrix `a.T`
  No need to copy the matrix and to create a new one

Some array creation routines

- numerical ranges: `arange`, `linspace`, `logspace`
- homogeneous data: `zeros`, `ones`
- diagonal elements: `diag`, `eye`
- random numbers: `rand`, `randint`

Numpy has an `append( )`-method. Avoid it if possible.

Your turn

Indexing and slicing in one dimension

1d arrays: indexing and slicing as for lists

- first element has index 0
- negative indices count from the end
- slices: `[start:stop:step]`
  without the element indexed by `stop`
- if values are omitted:
  - `start`: starting from first element
  - `stop`: until (and including) the last element
  - `step`: all elements between `start` and `stop`-1

Indexing and slicing in higher dimensions

- usual slicing syntax
- difference to lists:
  slices for the various axes separated by comma

a[2, -3]

0	1	2	3	4	5	6	7
8	9	10	11	12	13	14	15
16	17	18	19	20	21	22	23
24	25	26	27	28	29	30	31
32	33	34	35	36	37	38	39

Indexing and slicing in higher dimensions

- usual slicing syntax
- difference to lists:
  slices for the various axes separated by comma

a[:3, :5]

0	1	2	3	4	5	6	7
8	9	10	11	12	13	14	15
16	17	18	19	20	21	22	23
24	25	26	27	28	29	30	31
32	33	34	35	36	37	38	39

Indexing and slicing in higher dimensions

Your turn

0	1	2	3	4	5	6	7
8	9	10	11	12	13	14	15
16	17	18	19	20	21	22	23
24	25	26	27	28	29	30	31
32	33	34	35	36	37	38	39

Indexing and slicing in higher dimensions

a[-3:, -3:]

0	1	2	3	4	5	6	7
8	9	10	11	12	13	14	15
16	17	18	19	20	21	22	23
24	25	26	27	28	29	30	31
32	33	34	35	36	37	38	39

Indexing and slicing in higher dimensions

Your turn

0	1	2	3	4	5	6	7
8	9	10	11	12	13	14	15
16	17	18	19	20	21	22	23
24	25	26	27	28	29	30	31
32	33	34	35	36	37	38	39

Indexing and slicing in higher dimensions

Your turn

a[:, 3]

0	1	2	3	4	5	6	7
8	9	10	11	12	13	14	15
16	17	18	19	20	21	22	23
24	25	26	27	28	29	30	31
32	33	34	35	36	37	38	39

Indexing and slicing in higher dimensions

Your turn

0	1	2	3	4	5	6	7
8	9	10	11	12	13	14	15
16	17	18	19	20	21	22	23
24	25	26	27	28	29	30	31
32	33	34	35	36	37	38	39

Indexing and slicing in higher dimensions

a[1, 3:6]

0	1	2	3	4	5	6	7
8	9	10	11	12	13	14	15
16	17	18	19	20	21	22	23
24	25	26	27	28	29	30	31
32	33	34	35	36	37	38	39

Indexing and slicing in higher dimensions

Your turn

0	1	2	3	4	5	6	7
8	9	10	11	12	13	14	15
16	17	18	19	20	21	22	23
24	25	26	27	28	29	30	31
32	33	34	35	36	37	38	39

Indexing and slicing in higher dimensions

a[1::2, ::3]

0	1	2	3	4	5	6	7
8	9	10	11	12	13	14	15
16	17	18	19	20	21	22	23
24	25	26	27	28	29	30	31
32	33	34	35	36	37	38	39

Fancy indexing – Boolean mask

a[a % 3 == 0]

0	1	2	3	4	5	6	7
8	9	10	11	12	13	14	15
16	17	18	19	20	21	22	23
24	25	26	27	28	29	30	31
32	33	34	35	36	37	38	39

Fancy indexing – array of integers

a[(1, 1, 2, 2, 3, 3), (3, 4, 2, 5, 3, 4)]

0	1	2	3	4	5	6	7
8	9	10	11	12	13	14	15
16	17	18	19	20	21	22	23
24	25	26	27	28	29	30	31
32	33	34	35	36	37	38	39

Application: sieve of Eratosthenes

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49

0	1	**2**	3	~~4~~	5	~~6~~	7	~~8~~	9	~~10~~	11	~~12~~	13	~~14~~	15	~~16~~	17	~~18~~	19	~~20~~	21	~~22~~	23	~~24~~
25	~~26~~	27	~~28~~	29	~~30~~	31	~~32~~	33	~~34~~	35	~~36~~	37	~~38~~	39	~~40~~	41	~~42~~	43	~~44~~	45	~~46~~	47	~~48~~	49

0	1	**2**	**3**	~~4~~	5	~~6~~	7	~~8~~	~~9~~	~~10~~	11	~~12~~	13	~~14~~	~~15~~	~~16~~	17	~~18~~	19	~~20~~	~~21~~	~~22~~	23	~~24~~
25	~~26~~	~~27~~	~~28~~	29	~~30~~	31	~~32~~	~~33~~	~~34~~	35	~~36~~	37	~~38~~	~~39~~	~~40~~	41	~~42~~	43	~~44~~	~~45~~	~~46~~	47	~~48~~	49

0	1	**2**	**3**	~~4~~	**5**	~~6~~	7	~~8~~	~~9~~	~~10~~	11	~~12~~	13	~~14~~	~~15~~	~~16~~	17	~~18~~	19	~~20~~	~~21~~	~~22~~	23	~~24~~
~~25~~	~~26~~	~~27~~	~~28~~	29	~~30~~	31	~~32~~	~~33~~	~~34~~	~~35~~	~~36~~	37	~~38~~	~~39~~	~~40~~	41	~~42~~	43	~~44~~	~~45~~	~~46~~	47	~~48~~	49

0	1	**2**	**3**	~~4~~	**5**	~~6~~	**7**	~~8~~	~~9~~	~~10~~	11	~~12~~	13	~~14~~	~~15~~	~~16~~	17	~~18~~	19	~~20~~	~~21~~	~~22~~	23	~~24~~
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0	1	**2**	**3**	~~4~~	**5**	~~6~~	**7**	~~8~~	~~9~~	~~10~~	**11**	~~12~~	**13**	~~14~~	~~15~~	~~16~~	**17**	~~18~~	**19**	~~20~~	~~21~~	~~22~~	**23**	~~24~~
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Axes

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

axis 1 →

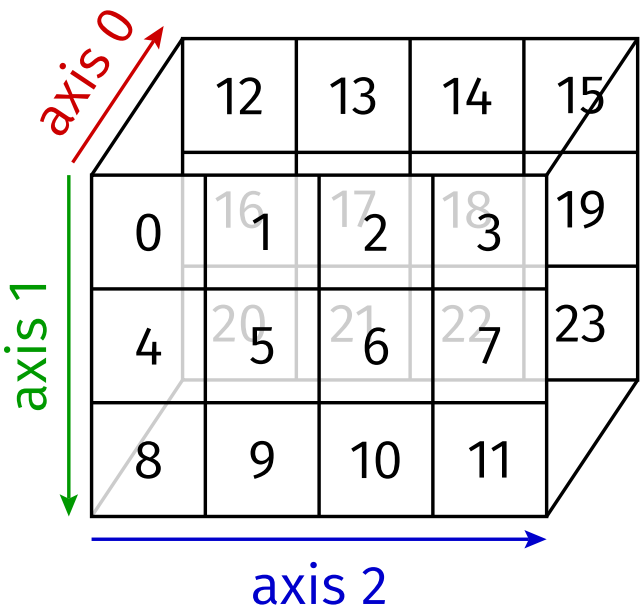
axis 0 ↓

$$\begin{pmatrix} a[0, 0] & a[0, 1] & a[0, 2] \\ a[1, 0] & a[1, 1] & a[1, 2] \\ a[2, 0] & a[2, 1] & a[2, 2] \end{pmatrix}$$

Your turn

```
np.sum(a)
np.sum(a, axis=...)
```

Axes in more than two dimensions




```
array([[[ 0,  1,  2,  3],
        [ 4,  5,  6,  7],
        [ 8,  9, 10, 11]],

       [[12, 13, 14, 15],
        [16, 17, 18, 19],
        [20, 21, 22, 23]]])
```

create this array and produce 2d arrays by cutting perpendicular to the axes 0, 1, and 2

Matrix multiplication

$$\begin{pmatrix} 0 & 1 \\ 2 & 3 \end{pmatrix}\begin{pmatrix} 4 & 5 \\ 6 & 7 \end{pmatrix} = \begin{pmatrix} \boxed{6} & 7 \\ 26 & 31 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 1 \\ 2 & 3 \end{pmatrix}\begin{pmatrix} 4 & 5 \\ 6 & 7 \end{pmatrix} = \begin{pmatrix} 6 & \boxed{7} \\ 26 & 31 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 1 \\ 2 & 3 \end{pmatrix}\begin{pmatrix} 4 & 5 \\ 6 & 7 \end{pmatrix} = \begin{pmatrix} 6 & 7 \\ \boxed{26} & 31 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 1 \\ 2 & 3 \end{pmatrix}\begin{pmatrix} 4 & 5 \\ 6 & 7 \end{pmatrix} = \begin{pmatrix} 6 & 7 \\ 26 & \boxed{31} \end{pmatrix}$$

try `np.dot(•, •)`

`•.dot(•)`

`• @ •` *)

*) Python ≥3.5, NumPy ≥1.10

Mathematical functions in NumPy

Universal functions (ufuncs) take ndarrays as argument

Trigonometric functions

sin, cos, tan, arcsin, arccos, arctan, hypot, arctan2, degrees, radians, unwrap, deg2rad, rad2deg

Hyperbolic functions

sinh, cosh, tanh, arcsinh, arccosh, arctanh

Rounding

around, round_, rint, fix, floor, ceil, trunc

Sums, products, differences

prod, sum, nansum, cumprod, cumsum, diff, ediff1d, gradient, cross, trapz

Exponents and logarithms

exp, expm1, exp2, log, log10, log2, log1p, logaddexp, logaddexp2

Other special functions

i0, sinc

Floating point routines

signbit, copysign, frexp, ldexp

Arithmetic operations

add, reciprocal, negative, multiply, divide, power, subtract, true_divide, floor_divide, fmod, mod, modf, remainder

Handling complex numbers

angle, real, imag, conj

Miscellaneous

convolve, clip, sqrt, square, absolute, fabs, sign, maximum, minimum, fmax, fmin, nan_to_num, real_if_close, interp

Many more special functions are provided as ufuncs by SciPy

Linear algebra in NumPy

```
import numpy.linalg as LA
```

Matrix and vector products

dot, vdot, inner, outer, matmul, tensordot, einsum, LA.matrix_power, kron

Decompositions

LA.cholesky, LA.qr, LA.svd

Matrix eigenvalues

LA.eig, LA.eigh, LA.eigvals, LA.eigvalsh

Norms and other numbers

LA.norm, LA.cond, LA.det, LA.matrix_rank, LA.slogdet, trace

Solving equations and inverting matrices LA.solve, LA.tensorsolve, LA.lstsq, LA.inv, LA.pinv, LA.tensorinv

hint: see also the methods for linear algebra in SciPy

Statistics in NumPy

Order statistics

amin, amax, nanmin, nanmax, ptp, percentile, nanpercentile

Averages and variances

median, average, mean, std, var, nanmedian, nanmean, nanstd, nanvar

Correlating

corrcoef, correlate, cov

Histograms

histogram, histogram2d, histogramdd, bincount, digitize

Sorting, searching, and counting in NumPy

Sorting

sort, lexsort, argsort, ndarray.sort, msort, sort_complex, partition, argpartition

Searching

argmax, nanargmax, argmin, nanargmin, argwhere, nonzero, flatnonzero, where, searchsorted, extract

Counting

count_nonzero

Application: identify entry closest to $1/2$

$$\begin{pmatrix} 0.05344164 & 0.37648768 & 0.80691163 & 0.71400815 \\ 0.60825034 & 0.35778938 & 0.37393356 & 0.32615374 \\ 0.83118547 & 0.33178711 & 0.21548027 & 0.42209291 \end{pmatrix}$$

$$\Downarrow$$

$$\begin{pmatrix} 0.37648768 \\ 0.60825034 \\ 0.42209291 \end{pmatrix}$$

hint: use `np.argsort`

Polynomials in NumPy

Power series: `numpy.polynomial.polynomial`

Polynomial Class

Polynomial

Basics

polyval, polyval2d, polyval3d, polygrid2d, polygrid3d, polyroots, polyfromroots

Fitting

polyfit, polyvander, polyvander2d, polyvander3d

Calculus

polyder, polyint

Algebra

polyadd, polysub, polymul, polymulx, polydiv, polypow

Miscellaneous

polycompanion, polydomain, polyzero, polyone, polyx, polytrim, polyline

also: Chebyshev, Legendre, Laguerre, Hermite polynomials

Some examples

```
P.Polynomial([24, -50, 35, -10, 1])
```

$$p_4(x) = x^4 - 10x^3 + 35x^2 - 50x + 24 = (x-1)(x-2)(x-3)(x-4)$$

```
p4.deriv()
```

$$\frac{\mathrm{d}p_4(x)}{\mathrm{d}x} = 4x^3 - 30x^2 + 70x - 50$$

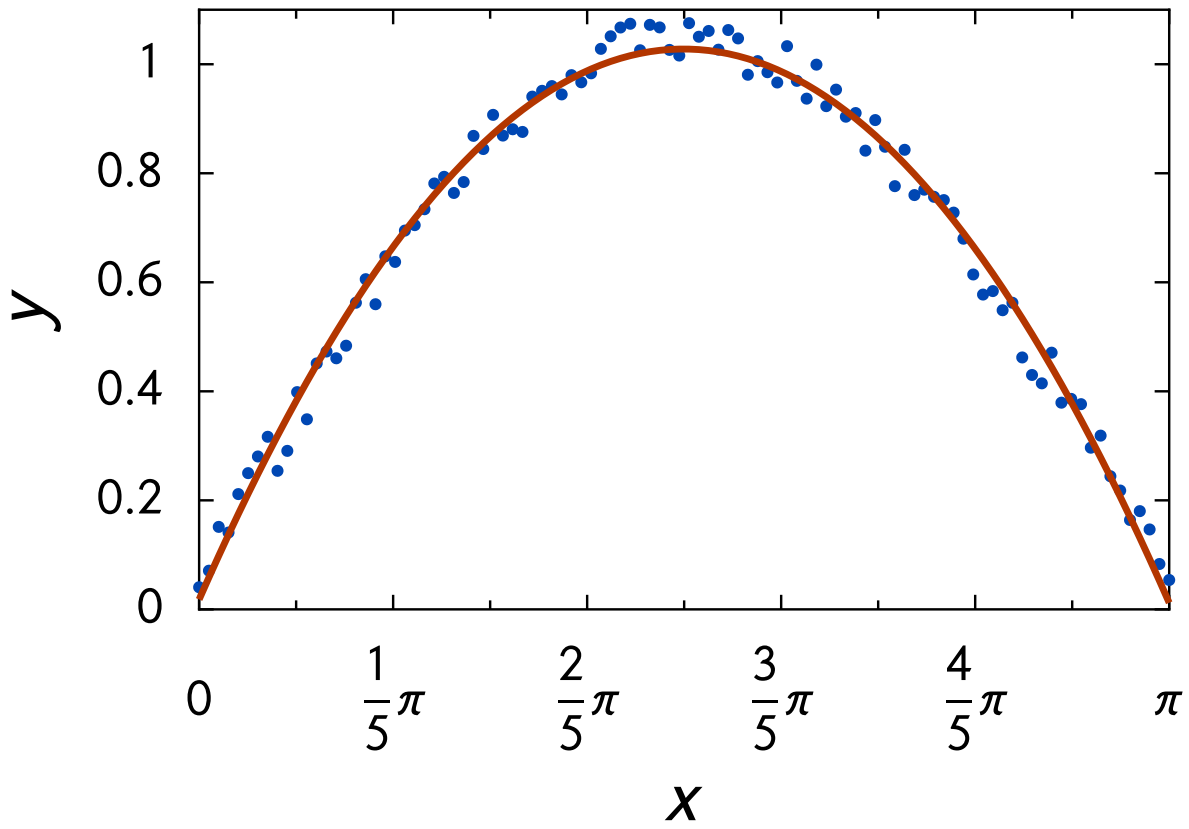
```
p4.integ()
```

$$\int p_4(x)\mathrm{d}x = \frac{1}{5}x^5 - \frac{5}{2}x^4 + \frac{35}{3}x^3 - 25x^2 + 24x + C$$

```
p4.polydiv()
```

$$\frac{p_4(x)}{2x+1} = \frac{1}{2}x^3 - \frac{21}{4}x^2 + \frac{161}{8}x - \frac{561}{16} + \frac{945}{16p_4(x)}$$

Application: polynomial fit



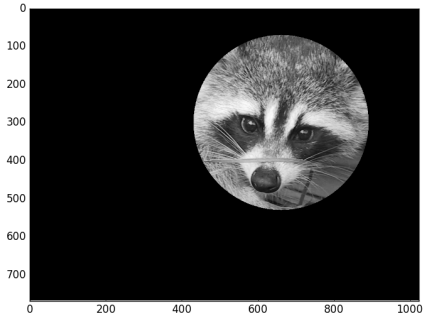
Your turn

add some noise to a function and fit it to a polynomial

see `scipy.optimize.curve_fit` for general fit functions

Application: image manipulation

```
from scipy import misc
face = misc.face(gray=True)
```



Your turn